ReconPilot

AI-assisted reconciliation break investigation and resolution for Indian financial institutions

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Raksha Kalegowda (2025MB26160) · Nicole Maria Gomes (2025MB26037) · Gurvinder Kaur (2025MB26010) · Achal Tiwari (2025MB26002) · Gokulkrishnan (2025MB26210) · Thirumanoor Venkata Phani Bhushan (2025MB26070)

Matching is solved; investigating breaks is not

- Breaks = items that disagree across internal ledger, bank statement and payment-gateway settlement
- Analysts compare three files field by field, recall rules and search past cases by memory
- Recurring root causes are rarely fixed upstream
- UPI, NEFT and IMPS volumes multiply the items each institution must reconcile

The decision we improve

"What caused this break, and what should I do about it?"

Evidence status

Problem framing is a team assumption. No customer interviews yet — discovery is phase 1.

Target users in Indian financial institutions

Reconciliation analyst

When a break appears in today's UPI/NEFT reconciliation, I want to see why and what fixed similar breaks, so I can resolve it correctly before cut-off.

Operations manager

When breaks pile up near month-end, I want to see high-risk and recurring breaks, so I can allocate analysts and push upstream fixes.

Risk / audit (secondary)

When I review a resolved break, I want the evidence and the approver, so I can confirm the control operated.

Competitors: strong incumbents, mid-market gap unvalidated

	Duco	SmartStream	Fractal Recon Assist	ReconPilot
Vendor AI claim	AI matching, label prediction, agentic workspace	"Affinity" observational-learning ML	Positions on faster resolution, accuracy, prevention	None implemented: rules + hand-weighted similarity
Focus	Large FIs, no-code cloud	Large banks, enterprise	Large enterprises	UPI/NEFT/IMPS, mid-size Indian FIs
Our angle	—	—	Near-identical positioning to ours	India-first formats, explainability, simulator + radar

Build the investigation layer; partner for matching and connectors. Vendor descriptions are from public trade-press and vendor pages, not product trials.

Product: one decision loop, five screens, human in control



- Break DNA: structured fingerprint (type, amount/date gaps, reference similarity, sources)
- Probable root cause with evidence and rule IDs; similar historical cases with outcomes
- Next best action with confidence; below 55 the suggestion becomes Investigate (analyst decides)
- No automatic resolution or posting — every decision needs a named analyst

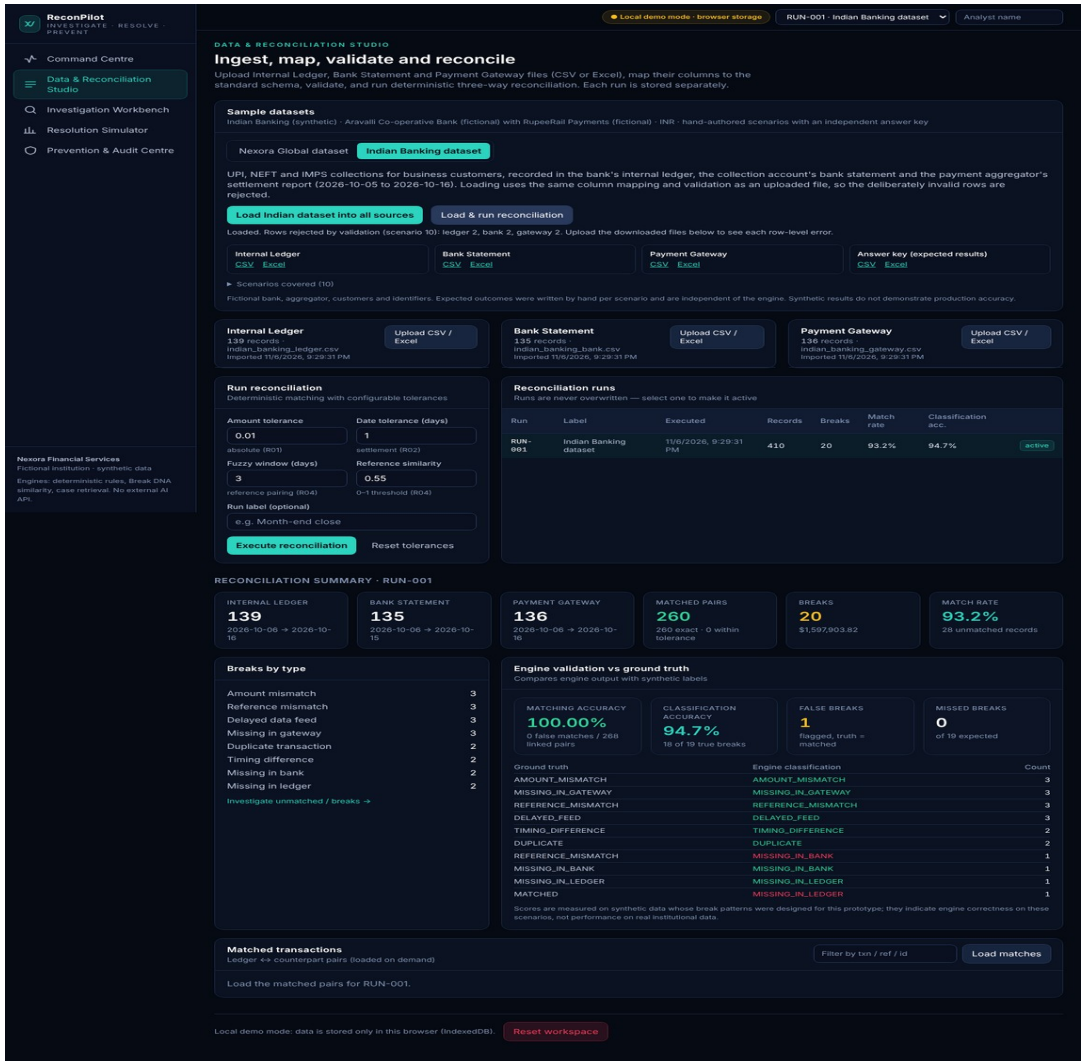
Architecture and the actual AI role

Capability	Technique today	Trained ML?	LLM?	Future
Matching & break detection	Deterministic rules (R01–R08, R28 currency guard)	No	No	Keep deterministic
Root cause	Hand-written rules (R09–R19, R28)	No	No	Learned classifier vs rule baseline
Break DNA & similar cases	Hand-weighted feature similarity + nearest-case retrieval	No	No	Learned weights
Next best action	Rule map; history override only under R24	No	No	Calibrated ranking
Confidence	Additive evidence points, uncalibrated	No	No	Calibration on outcomes

Next.js 16 + TypeScript · engines run in browser and API routes · IndexedDB or Supabase · Vercel · 28 rules · no external AI API

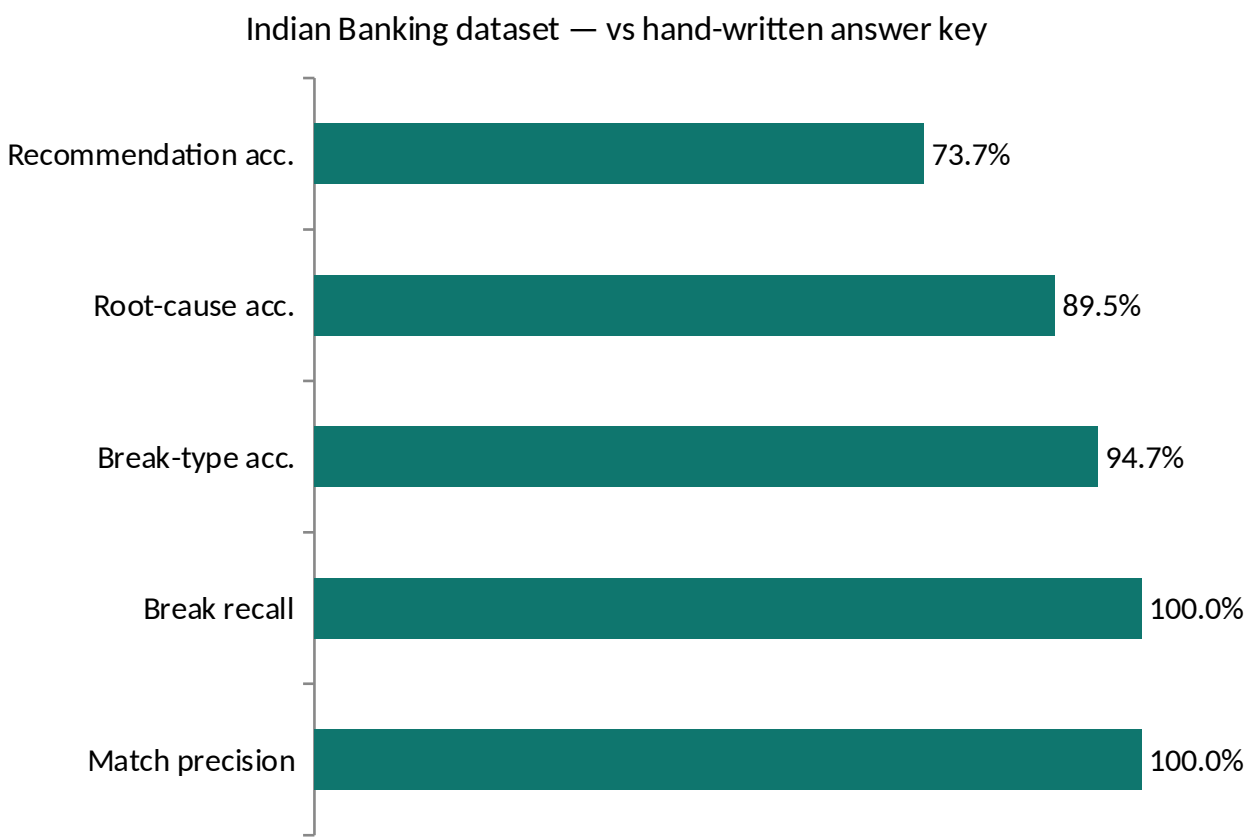
"Learning" = recorded outcomes are retrieved later. Nothing is retrained.

Walkthrough: Indian Banking demo in five minutes



- Studio → Sample datasets → Indian Banking (fictional Aravalli Co-op Bank + RupeeRail aggregator, INR)
- Load & run: 260 matched pairs, 20 breaks; 6 invalid rows rejected with row-level errors
- Workbench: open the gateway fee-netting break (MDR + GST) and the transposed IMPS reference
- Simulator → approve with analyst name → audit log and Recurring Break Radar
- Nexora Global multi-currency dataset remains unchanged alongside

Evaluation: strong on designed cases, misses kept visible



Known failures (kept visible)

Paise rounding read as FX rounding (R10, conf. 90) · bank-substituted reference → false "missing in ledger" · 5 action misses (14/19): paise case, 2 R24 history overrides, 2 R23 low-confidence Investigate

Not real-world accuracy

Team-written scenarios (not blind). Nexora reported separately: 100% matching, 97.4–98.5% classification. 57 tests pass.

Business model: illustrative ROI, every input assumed

Scenario (assumed)	Breaks/mo	Min/break	₹/hr	Time saved	Op. cost/mo	Net saving/mo	Break-even
A · Small co-op bank	2,000	20	500	20%	₹45,000	₹21,667	13.5%
B · NBFC / mid-size bank	10,000	20	650	25%	₹1,00,000	₹4,41,667	4.6%
C · Large payments ops	40,000	15	800	30%	₹2,40,000	₹21,60,000	3.0%

- Monetisation (assumed): annual SaaS tiered by break volume + implementation fee
- Main cost: rule upkeep and support; model/API cost today ₹0 (no external AI)
- Freed hours are capacity, not cash. Not customer savings. Small co-op case is marginal → mid-size first

Risks, controls and launch blockers

Main risks

Wrong recommendation accepted → misstatement or delayed customer refund

Over-trust in uncalibrated confidence

INR rated riskier by currency-agnostic thresholds (R20)

History overrides from synthetic outcomes (R24)

Controls in place

Human approval for every decision (R25)

Low-confidence Investigate suggestion (R23)

Currency guard (R28) and row-level validation

Audit log of imports, runs, decisions — not tamper-proof

Launch blockers

No authentication in the cloud app — real financial data prohibited

No tamper-evident audit

Per-currency thresholds and domestic-rounding rule missing

xlsx 0.18.5 advisory; no security review

Roadmap with measurable proof gates

1

Discovery · 4–6 wks

≥6/10 confirm investigation is the main time sink; ≥2 shadow-pilot partners

2

Pilot readiness · 6–8 wks

Auth + roles, four-eyes, INR thresholds, tamper-evident audit; no high security findings

3

Shadow pilot · 8–12 wks

Root-cause acc. ≥80%; median time −20%; false resolutions not worse

4

Learned models · after pilot

Beat rule baseline by ≥5 pts on held-out data; calibrated confidence

Recommendation: PILOT preparation — not production

- Proven: the investigation workflow, explainable logic, human-in-the-loop control and an independent evaluation method on two synthetic datasets
- Not proven: time savings, real-world accuracy, buyer demand vs Duco / SmartStream / Fractal
- Next investment buys evidence: discovery interviews and a shadow pilot on anonymised data, after closing the launch blockers

Repo: github.com/krish19098/Reconpilot · Live: reconpilot-omega.vercel.app